MTM4692 - APPLIED SQL

MILESTONE 1: PROJECT PROPOSAL & TECHNICAL JUSTIFICATION

Course: MTM4692 - Applied SQL

Team Number: 19

GitHub Repository: https://github.com/betulfeyza/sql-project

Link of the video: https://stdyildizedu-my.sharepoint.com/:v:/g/personal/suzal_demirciler_std_yildiz_edu_tr/IQDCHvqiB_U0QZBDjrUWiDwMAaYYzCdTcETv8yY2b67E-zo?e=fJUCU7

Team Members:

Muhammed Emin Beyatlı

Ümmügülsüm Suzal Demirciler

Mehmet Barış Saç

Zeynep Cansu Ağbulut

Betül Feyza Çoker

1. Problem Statement

The Faculty of Chemical and Metallurgical Engineering (KMF) lacks a centralized digital system to monitor and manage classroom occupancy, scheduling, and temporary event reservations. Currently, students face difficulty finding available study spaces and classrooms with suitable technical equipment such as projectors, smart boards, or sufficient power outlets. In most cases, they rely on physically checking rooms, which wastes time and causes inefficiency.

At the same time, academic staff manage lecture schedules, exam planning, and event requests manually. This creates a high risk of classroom conflicts, scheduling overlaps, and inefficient allocation of faculty resources. Therefore, the faculty needs an integrated system that can support both daily operational needs and data-driven planning.

2. Project Justification & Applied SQL Vision

This project, titled "KMF Smart Classroom & Event Management System," aims to digitalize faculty resource management through an Applied SQL approach using a Role-Based Access Control (RBAC) structure implemented on top of a normalized SQLite database.

For Students:
The system provides a privacy-safe interface where students can view live classroom availability through a room-status screen supported by `v_student_live_status`. They can also use smart filters to search for rooms according to equipment features such as projector availability, smart board support, and the number of power outlets.

For Academics:
The system supports academic scheduling and decision-making by enabling academics to review exam coordination data, monitor pending reservation requests, and approve or reject student event

requests. Classroom equipment is stored in a JSON-based `specs` field, allowing flexible querying without changing the static schema.

The project reflects the goals of the Applied SQL course because SQL is not used only for storage, but also for enforcing business rules, supporting analytics, and powering the interface logic.

3. Preliminary Database Schema

The system is designed with a relational structure consisting of 6 core tables in compliance with 3NF normalization and data integrity principles:

Users: Stores user identity, department, email, and role information (`Student` / `Academic`) for role-based access.

Departments: Contains the five main departments of KMF:

- Mathematical Engineering
- Chemical Engineering
- Metallurgical and Materials Engineering
- Food Engineering
- Bioengineering

Classrooms: Stores classroom metadata such as room code, capacity, block, floor, and a JSON-based `specs` field for technical equipment tracking.

Academic_Schedules: Stores lecture, exam, and seminar schedules together with room, academic, department, and time information.

Event_Requests: Manages reservation requests submitted by users, including workflow states such as `Pending`, `Approved`, and `Rejected`, and tracks the academic approver when relevant.

Usage_Logs: Stores timestamp-based occupancy observations for classrooms, including room status and occupancy count for live monitoring and analytical reporting.

4. Proposed Advanced SQL Features

To fulfill the requirements of the Applied SQL curriculum, the following technical features are implemented:

Views:

- `v_student_live_status` for privacy-safe student access to live room information
- `v_exam_coordination` for academic reporting on exam planning, occupancy trends, and overlapping requests

Recursive CTE:
Used to generate a hierarchical faculty layout in the form of Block > Floor > Room.

Window Functions:
Used with `RANK()` or `DENSE_RANK()` to analyze and rank classrooms according to weekly utilization intensity.

JSON Support:
The `specs` column in `Classrooms` stores dynamic equipment information such as projector, power outlets, smart board, and air conditioning. This allows flexible querying without schema redesign.

Triggers & Constraints:
Business logic is enforced at SQL level, including:

- only academic users can approve requests
- occupancy count cannot exceed room capacity
- classroom schedules cannot overlap
- pending or approved event requests cannot conflict with academic schedules
- approved event requests cannot overlap with other approved requests

5. Sample Analytical Queries

Query A: Finding Available Study Space (Using Views)

Objective: Find available classrooms in Block B with at least 10 power outlets.

```
SELECT room_code, capacity
FROM v_student_live_status
WHERE live_status = 'Available'
  AND block = 'B'
  AND power_outlets >= 10;
```

Query B: Utilization Ranking (Using Window Functions)

Objective: Rank classrooms by weekly utilization ratio based on occupancy logs.

```
WITH weekly_usage AS (
    SELECT
        c.room_id,
        c.room_code,
        ROUND(AVG(1.0 * ul.occupancy_count / c.capacity), 4) AS
avg_utilization_ratio
    FROM Classrooms c
    JOIN Usage_Logs ul ON ul.room_id = c.room_id
    WHERE datetime(ul.observed_at) >= datetime('now', '-7 days')
    GROUP BY c.room_id, c.room_code
)
SELECT
    room_code,
    avg_utilization_ratio,
    RANK() OVER (ORDER BY avg_utilization_ratio DESC) AS utilization_rank
FROM weekly_usage;
```